

Article

S-PACE: Sync Physiological Alignment Conditioning Embedding for Robust Multimodal Emotion Recognition

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Abstract

Multimodal emotion recognition faces two persistent challenges when integrating physiological signals with behavioral modalities: (1) temporal misalignment arising from the inherent latency difference between fast behavioral responses and slower autonomic physiological reactions, and (2) reliability degradation caused by noisy or missing modalities under naturalistic conditions. To address these challenges, we propose SGMT (S-PACE Gated Multimodal Transformer), a unified architecture that performs Sync Physiological Alignment Conditioning Embedding (S-PACE), quality-aware multimodal fusion, and hierarchical temporal modeling. S-PACE aligns static physiological representations with time-indexed behavioral cues through cross-attentional conditioning, mitigating physiological-behavioral lag and enabling coherent multi-modal temporal integration. A Quality-Aware Gate dynamically modulates modality contributions based on estimated signal reliability, suppressing corrupted or absent modality streams. Aligned time-series features are processed by a Temporal Swin Transformer, and a Perceiver Fusion module integrates multi-scale temporal tokens into a compact emotion embedding, jointly predicting binary arousal-valence states and quadrant labels. We evaluate SGMT on two Korean naturalistic emotion corpora with complementary characteristics: KEMDy20 (audio + bio, 3-fold LOPPO) and K-EmoCon (audio + video + bio, 6-fold CV). On KEMDy20, SGMT achieves arousal UAR of 68.4% and valence UAR of 57.6%, with quadrant accuracy of 44.7% substantially exceeding chance level (25%). On K-EmoCon, SGMT attains arousal UAR of 62.9% and arousal F1 of 73.5%. Ablation studies confirm that each proposed component—S-PACE alignment, quality-aware gating, and multimodal fusion—contributes to overall performance, with the full model achieving more balanced arousal-valence recognition than unimodal baselines and outperforming simple concatenation fusion. Gate weight analysis reveals that the model automatically concentrates on video (~95%) when facial data is available and shifts to audio (~90%) when it is absent, suggesting adaptive modality reweighting behavior depending on modality availability.

Keywords: multimodal emotion recognition; physiological signal alignment; quality-aware fusion; temporal transformer; bio-behavioral synchronization; affective computing; gated fusion; wearable biosensors

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1. Introduction

Emotions constitute complex physiological, cognitive, and behavioral responses elicited by internal or external stimuli. Automatic emotion recognition has emerged as a critical research area [1], with applications spanning driver safety monitoring [2],

healthcare [3], human–computer interaction [4], and affective computing benchmarks [5]. Traditional approaches predominantly rely on observable behavioral cues such as facial expressions and speech prosody. While effective in controlled settings, these signals can be consciously modulated or suppressed, limiting their reliability for naturalistic emotion assessment.

Physiological signals—electrodermal activity (EDA), blood volume pulse (BVP), skin temperature (TEMP), and heart-rate variability (HRV)—offer a complementary perspective. Generated by the autonomic nervous system, these responses are largely involuntary and thus provide more objective indicators of affective states [6]. Recent advances in wearable sensing technology have made peripheral physiological signals increasingly accessible, motivating research on integrating them with behavioral modalities for robust emotion recognition [7].

However, effectively integrating physiological signals into multimodal emotion recognition introduces two fundamental challenges that remain insufficiently addressed by existing architectures:

(1) Temporal misalignment between behavioral and physiological responses. When a person experiences an emotional stimulus, behavioral responses (facial expressions, vocal changes) manifest within milliseconds, whereas physiological responses (EDA peaks, heart-rate changes) exhibit latencies of several seconds [8]. This bio-behavioral temporal lag means that naively fusing time-aligned features from different modalities conflates causally unrelated signals, degrading recognition performance.

(2) Modality-specific noise and availability. In naturalistic settings, individual modalities frequently suffer from corruption: facial video may be occluded, speech segments may contain background noise, and biosensor signals may include motion artifacts. Moreover, some modalities may be entirely unavailable—for instance, facial video is absent in telephone-based or audio-only scenarios. A robust fusion strategy must dynamically adjust modality contributions based on signal quality rather than treating all inputs equally.

To address these challenges, we propose **SGMT** (S-PACE Gated Multimodal Transformer), a unified architecture built around the principle of *bio-anchored alignment*. Unlike conventional multimodal emotion recognition systems that treat all modalities symmetrically, SGMT positions physiological signals as the *anchor modality* and behavioral signals (audio, video) as *conditioning modalities*. The rationale for this asymmetric design is twofold: physiological signals are (i) involuntary and thus less susceptible to social masking, and (ii) present across virtually all sensing configurations, including audio-only or video-occluded scenarios, making them a stable reference for fusion. The core component is **S-PACE** (Stimulus-Phase Alignment via Cross-modal Embedding), a cross-attention mechanism where behavioral features serve as queries attending over physiological sensor channels. This produces time-aligned bio representations that mitigate the physiological–behavioral temporal lag at the representation level.

SGMT further incorporates a **Quality-Aware Gate** that estimates per-timestep modality reliability from signal quality proxies and produces soft fusion weights via temperature-annealed softmax. This mechanism enables the model to automatically suppress corrupted modalities and concentrate on reliable signals. A **hierarchical Temporal Swin Transformer** captures multi-scale temporal dynamics through windowed self-attention with progressive resolution reduction, and a **Perceiver Fusion** module compresses heterogeneous multimodal tokens into a fixed-dimensional latent representation for downstream prediction.

We evaluate SGMT on two Korean naturalistic emotion corpora chosen for their complementary modality configurations: **KEMDy20** [9], which provides audio and bio signals without facial video in dyadic conversation settings, and **K-EmoCon** [10], which offers all three modalities (audio, video, bio) in a paired conversation paradigm. This

cross-dataset evaluation is specifically designed to test whether the same architecture can adapt to different modality availability scenarios through the gating mechanism, rather than requiring separate model variants for each configuration.

The main contributions of this study are summarized as follows:

- We propose **S-PACE**, a cross-attentional alignment module that mitigates physiological-behavioral temporal lag by using behavioral features as queries over bio-sensor channels, establishing physiological signals as the anchor modality for multimodal fusion.
- We introduce a **Quality-Aware Gate** with temperature annealing that dynamically adjusts modality contributions based on estimated signal reliability, enabling graceful degradation when modalities are noisy or absent.
- We design a **unified architecture** combining hierarchical Temporal Swin Transformers and Perceiver Fusion that generalizes across different modality configurations without requiring architectural modifications.
- We conduct experiments on two naturalistic Korean emotion datasets with complementary modality availability, showing that the full multimodal architecture with S-PACE achieves more balanced arousal–valence recognition than unimodal baselines, and gate weight analysis suggests that the model learns adaptive modality selection without explicit supervision.

2. Related Works

2.1. Audio–Visual Emotion Recognition

Audio–visual emotion recognition has been extensively studied as the two most accessible behavioral modalities. Early approaches employed handcrafted features such as Mel-frequency cepstral coefficients (MFCCs) for audio and action units (AUs) or facial landmarks for video, combined through early or late fusion [11]. With the advent of deep learning, end-to-end architectures became dominant. Attention-based fusion mechanisms [12,13] learn to weight temporal segments differentially, while cross-modal transformers [14] enable modality interaction through cross-attention layers.

Recent work has explored self-supervised pre-training for audio–visual representations. The Audio Spectrogram Transformer (AST) [15] demonstrated that vision transformer architectures pre-trained on large-scale audio data achieve strong performance on downstream tasks. Similarly, face analysis has benefited from large-scale pre-trained visual backbones [16]. However, audio–visual approaches inherently lack physiological grounding, making them susceptible to deliberate emotional suppression or social display rules.

2.2. Physiological Signal-Based Emotion Recognition

Physiological signals provide involuntary indicators of emotional states. EDA reflects sympathetic nervous system arousal, BVP and inter-beat interval (IBI) capture cardiac dynamics, and skin temperature correlates with peripheral blood flow changes during emotional episodes [6]. Several benchmark datasets have been established for physiological emotion recognition, including DEAP [17] and AMIGOS [18], which provide multi-channel biosignals with emotion annotations.

Traditional approaches extract hand-crafted features (peak amplitude, rise time, spectral power) and apply classifiers such as SVMs or random forests [19]. Deep learning approaches have increasingly adopted convolutional networks applied to spectrogram, recurrence plot, or Gramian Angular Field representations of physiological time series [20–22]. The K-EmoCon dataset [10] introduced a benchmark for multimodal emotion recognition combining peripheral biosignals with audio and video in naturalistic conversation settings.

A key challenge in physiological emotion recognition is the *temporal lag* between stimulus presentation and physiological response. EDA responses typically peak 1–5 seconds after stimulus onset [23], while cardiovascular responses exhibit variable latencies depending on the autonomic pathway involved. This temporal misalignment has been largely ignored in prior fusion architectures, which assume synchronized feature extraction across modalities.

2.3. Multimodal Fusion Strategies

Multimodal fusion strategies can be categorized along two dimensions: *when* fusion occurs (early, intermediate, late) and *how* modality interactions are modeled [24].

Early fusion concatenates raw or low-level features before processing [25], preserving fine-grained cross-modal correlations but requiring all modalities to share a common temporal resolution. **Late fusion** combines modality-specific predictions, offering robustness to missing modalities but sacrificing cross-modal interaction [26]. **Intermediate fusion** approaches, which fuse at the representation level, have gained popularity through attention-based mechanisms [27]. Recent methods such as MISA [28] learn modality-invariant and modality-specific subspaces to disentangle shared and private information across modalities.

Gated fusion mechanisms learn modality-specific weights to control information flow. Gated Multimodal Units [29] use sigmoid gates to regulate each modality's contribution. More recently, quality-aware approaches incorporate external reliability estimates into the gating mechanism [30], though most focus on audio–visual modalities without physiological signals.

The Perceiver architecture [31] introduced a powerful framework for handling heterogeneous multi-modal inputs through cross-attention with learnable latent tokens, enabling fixed-dimensional representations regardless of input modality count or sequence length.

Recent studies on the K-EmoCon benchmark illustrate the diversity of approaches and task formulations in this domain. Yu et al. [32] proposed a hierarchical gated residual network operating exclusively on peripheral physiological signals, achieving strong performance through recursive feature aggregation across K-EmoCon, DEAP, and ILEmo with LOSO validation. Song et al. [33] introduced speaker diarization as a preprocessing step to isolate single-speaker audio before multimodal fusion with physiological signals. Alhussein et al. [34] reframed the task as *emotion climate* classification using bispectral audio features, defining a collective-level affective construct distinct from per-subject emotion recognition. Kim and Hong [35] integrated environmental and personality metadata with physiological signals through a Swin Transformer feature extractor for continuous emotion intensity prediction. These studies highlight that accepted approaches span a wide range of task formulations (binary classification, emotion climate, continuous regression), modality combinations, and evaluation protocols, making direct cross-study comparison inherently challenging.

Our work differs from prior approaches in three key aspects: (1) we explicitly model the physiological–behavioral temporal lag through cross-attentional alignment rather than assuming temporal synchrony, addressing a well-documented but architecturally neglected challenge in biosignal-based emotion recognition; (2) our quality-aware gate incorporates batch-normalized signal quality proxies with temperature annealing, enabling a smooth transition from uniform exploration to specialized modality selection during training; and (3) we adopt an asymmetric bio-anchored fusion design in which physiological signals serve as a stable reference modality, in contrast to the symmetric treatment of modalities prevalent in prior architectures.

3. Methodology

3.1. Overview

Figure 1 illustrates the overall architecture of SGMT. The design follows a *bio-anchored* principle: physiological signals occupy the central processing axis, while behavioral signals (video, audio) serve as conditioning inputs. The forward pass consists of seven stages: (1) modality-specific encoding, (2) positional and type embedding, (3) S-PACE bio-behavioral alignment, (4) quality-aware gated fusion, (5) hierarchical temporal processing, (6) perceiver cross-modal fusion, and (7) multi-task prediction.

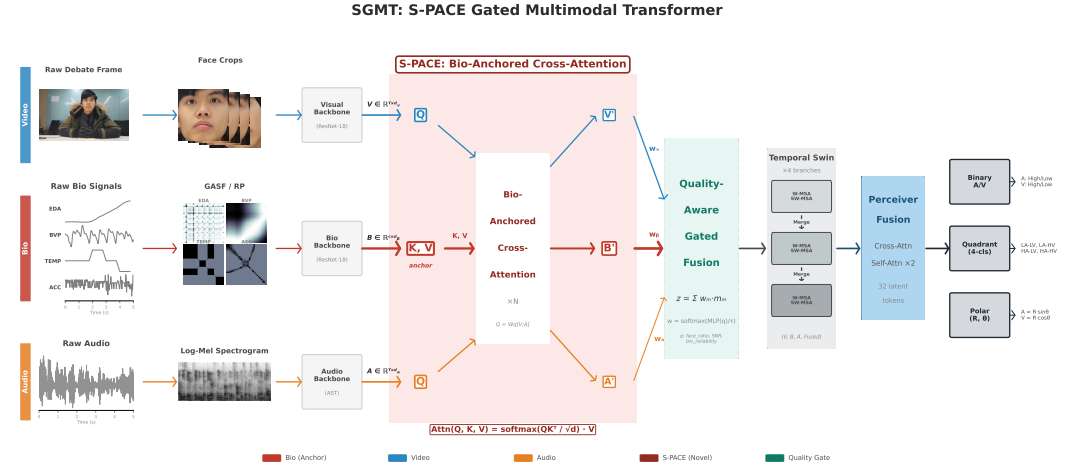


Figure 1. End-to-end architecture of the proposed SGMT model with real K-EmoCon data examples. Raw inputs (left) are processed by modality-specific encoders. Physiological signals (center, red) serve as the anchor modality with Key and Value roles in the Bio-Anchored Cross-Attention (S-PACE), while video (top, blue) and audio (bottom, orange) provide conditioning Queries. The Quality-Aware Gate dynamically weights modality contributions (w_v , w_b , w_a) based on signal reliability, followed by hierarchical Temporal Swin processing and Perceiver Fusion for multi-task prediction.

3.2. Modality-Specific Encoders

3.2.1. Bio-Sensor Encoder

The Bio-Sensor Encoder processes C -channel physiological signal images independently through per-channel ResNet-18 [36] backbones. Each biosensor channel is converted to a single-channel 2D time-series image using a transformation matched to its signal characteristics. For oscillatory signals (EDA, BVP), we apply the Gramian Angular Summation Field (GASF) [21], which encodes pairwise temporal correlations as $\text{GASF}(i, j) = \cos(\phi_i + \phi_j)$ where $\phi = \arccos(\tilde{x})$ is the angular representation of the min-max normalized signal. For signals with low-frequency dynamics (TEMP, ACC), we use a binary recurrence plot $\text{RP}(i, j) = \mathbb{I}[|x_i - x_j| < \epsilon]$ [22], which captures state-transition patterns more effectively than GASF for signals with limited oscillatory variation. Each sensor produces a single-channel image and is encoded as:

$$\mathbf{b}_c = \text{Linear}(\text{ResNet18}(\mathbf{x}_c^{\text{bio}})), \quad c = 1, \dots, C \quad (1)$$

where $\mathbf{x}_c^{\text{bio}} \in \mathbb{R}^{1 \times H \times W}$ is the single-channel image and $\mathbf{b}_c \in \mathbb{R}^d$. The resulting tensor $\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_C] \in \mathbb{R}^{C \times d}$ provides C sensor-specific tokens. We use $C=4$ for K-EmoCon (BVP, EDA, TEMP, ACC) and $C=3$ for KEMDy20 (EDA, TEMP, IBI).

3.2.2. Video Encoder

The Video Encoder processes T face crop frames through a shared ResNet-18 backbone:

$$\mathbf{v}_t = \text{Linear}(\text{ResNet18}(\mathbf{x}_t^{\text{vid}})), \quad t = 1, \dots, T \quad (2)$$

where $\mathbf{x}_t^{\text{vid}} \in \mathbb{R}^{3 \times 48 \times 48}$ and $\mathbf{V} \in \mathbb{R}^{T \times d}$. When facial video is unavailable (e.g., KEMDy20), we set $\mathbf{V} = \mathbf{0}$ without architectural changes.

3.2.3. Audio Encoder

The Audio Encoder leverages a pre-trained Audio Spectrogram Transformer (AST) [15]:

$$\mathbf{A} = \text{MLP}(\text{Pool}_T(\text{AST}(\mathbf{x}^{\text{aud}}))) \quad (3)$$

where $\mathbf{x}^{\text{aud}} \in \mathbb{R}^{1 \times 64 \times T_{\text{mel}}}$ is the log-Mel spectrogram, AST produces patch-level features, Pool_T is an adaptive average pooling to $T=16$ temporal steps, and MLP projects to d dimensions. The AST backbone is initially frozen and unfrozen at epoch 6 for fine-tuning.

3.3. Positional and Type Embeddings

Each modality token receives three additive embeddings:

$$\hat{\mathbf{m}}_t = \mathbf{m}_t + \mathbf{e}_t^{\text{pos}} + \mathbf{e}_k^{\text{type}} + \mathbf{e}_s^{\text{spk}} \quad (4)$$

where $\mathbf{e}_t^{\text{pos}}$ is a learnable temporal position embedding ($t = 1, \dots, T$), $\mathbf{e}_k^{\text{type}}$ distinguishes modality type ($k \in \{0:\text{bio}, 1:\text{audio}, 2:\text{video}\}$), and $\mathbf{e}_s^{\text{spk}}$ encodes speaker identity ($s \in \{0, 1\}$) for dyadic conversation modeling.

3.4. S-PACE: Stimulus-Phase Alignment via Cross-modal Embedding

S-PACE is the core alignment component designed to mitigate the temporal misalignment between physiological and behavioral signals. The key design principle is that *behavioral signals should query into physiological signals*, not the reverse—because the question being asked at each timestep is: “Which physiological sensor channels are most relevant to the current behavioral observation?” This formulation is motivated by the fact that behavioral responses precede physiological reactions temporally; hence, using behavioral features as queries allows the model to learn which physiological channels are causally related to the observed behavioral state at each time step.

Formally, S-PACE performs scaled dot-product cross-attention [12] where behavioral features form the queries and bio-sensor tokens serve as keys and values:

$$\mathbf{Q} = \mathbf{W}_Q[\hat{\mathbf{V}}; \hat{\mathbf{A}}] \in \mathbb{R}^{T \times d} \quad (5)$$

$$\mathbf{K} = \mathbf{W}_K \mathbf{B} \in \mathbb{R}^{C \times d} \quad (6)$$

$$\mathbf{V}_{\text{att}} = \mathbf{W}_V \mathbf{B} \in \mathbb{R}^{C \times d} \quad (7)$$

$$\mathbf{B}_t = \text{MLP}\left(\text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right)\mathbf{V}_{\text{att}}\right) \in \mathbb{R}^{T \times d} \quad (8)$$

Here, $[\hat{\mathbf{V}}; \hat{\mathbf{A}}]$ denotes the concatenation of video and audio features along the channel dimension, and $\mathbf{B}_t$ is the *time-aligned bio representation*. The attention matrix $\alpha \in \mathbb{R}^{T \times C}$ captures which bio-sensor channel is most relevant at each temporal step—effectively learning a soft temporal alignment between the fast behavioral timeline and the slower physiological response.

When video is absent (skip-video mode), $\hat{\mathbf{V}} = \mathbf{0}$, and the query reduces to audio-only conditioning. This graceful degradation is a direct consequence of the bio-anchored design: the anchor modality (bio) always provides keys and values, while the conditioning signal adapts to whatever behavioral information is available.

3.5. Quality-Aware Gated Fusion

After S-PACE alignment, three modality streams—video $\hat{\mathbf{V}}$, audio $\hat{\mathbf{A}}$, and aligned bio $\mathbf{B}_t$ —must be fused into a single representation. Rather than fixed or attention-derived weights, we propose a Quality-Aware Gate that incorporates external signal quality estimates.

Let $\mathbf{q} = [q_{\text{face}}, q_{\text{audio}}, q_{\text{bio}}] \in \mathbb{R}^3$ denote quality proxies computed from the input signals. Specifically, $q_{\text{face}} = \frac{1}{T} \sum_{t=1}^T \mathbb{1}[\text{face detected at } t]$ measures the proportion of frames with valid face detections; $q_{\text{audio}} = 1 - \frac{\bar{s}}{s_{\text{max}}}$ is a simplified spectral flatness proxy computed from the log-Mel spectrogram, where lower flatness (i.e., higher q_{audio}) indicates the presence of structured speech energy; and $q_{\text{bio}} = \frac{1}{C} \sum_{c=1}^C (\max(\mathbf{x}_c^{\text{bio}}) - \min(\mathbf{x}_c^{\text{bio}}))$ captures the mean dynamic range across bio-sensor channels, with higher values indicating active physiological variation rather than sensor malfunction. We first apply batch-wise z-score normalization for scale invariance:

$$\tilde{\mathbf{q}} = \frac{\mathbf{q} - \mu_{\mathcal{B}}}{\sigma_{\mathcal{B}} + \epsilon} \quad (9)$$

The gate weight is computed as:

$$\mathbf{w}_t = \text{softmax}\left(\frac{\text{MLP}([\hat{\mathbf{V}}_t; \hat{\mathbf{A}}_t; \mathbf{B}_{t,t}; \tilde{\mathbf{q}}])}{\tau}\right) \in \mathbb{R}^3 \quad (10)$$

$$\mathbf{z}_t = w_{t,1} \cdot \hat{\mathbf{V}}_t + w_{t,2} \cdot \hat{\mathbf{A}}_t + w_{t,3} \cdot \mathbf{B}_{t,t} \quad (11)$$

The temperature τ follows a cosine annealing schedule from $\tau_0=2.0$ to $\tau_1=1.0$ over training epochs. High initial temperature encourages uniform exploration across modalities, while decreasing temperature enables sharper specialization as the model learns modality reliability patterns.

3.6. Hierarchical Temporal Swin Transformer

Temporal dynamics are modeled through a 1D adaptation of the Swin Transformer [37], applied both to the fused representation and to each intra-modal stream independently.

Each Temporal Swin Encoder consists of three stages with progressive temporal resolution reduction:

- **Stage 0** ($T=16 \rightarrow 8$): Window self-attention blocks with window size $w=4$ and alternating shifted windows, followed by temporal merge (concatenation of adjacent pairs + linear projection).
- **Stage 1** ($T=8 \rightarrow 4$): Same architecture, operating at half temporal resolution.
- **Stage 2** ($T=4$): Final refinement without further downsampling.

Each stage contains D blocks alternating between window multi-head self-attention (W-MSA) and shifted window MSA (SW-MSA), with layer normalization [38] applied before each attention and feed-forward sub-layer:

$$\hat{\mathbf{x}}^l = \text{W-MSA}(\text{LN}(\mathbf{x}^{l-1})) + \mathbf{x}^{l-1} \quad (12)$$

$$\mathbf{x}^l = \text{FFN}(\text{LN}(\hat{\mathbf{x}}^l)) + \hat{\mathbf{x}}^l \quad (13)$$

We apply four parallel Temporal Swin Encoders: one post-gate encoder with depth (1, 1, 1) for the fused token $\mathbf{z}$, and three intra-modal encoders with depth (2, 2, 2) for video, audio, and bio streams independently. Face mask propagation ensures that invalid video frames are properly masked throughout the temporal hierarchy.

3.7. Perceiver Cross-Modal Fusion

All temporal tokens are concatenated and compressed through a Perceiver [31] module:

$$\mathbf{T} = [\mathbf{z}'; \mathbf{V}'; \mathbf{A}'; \mathbf{B}'] \in \mathbb{R}^{N \times d} \quad (14)$$

where primes denote post-Swin tokens ($N \approx 16$ total tokens after temporal merging).

A set of $L=32$ learnable latent tokens $\mathbf{\Lambda} \in \mathbb{R}^{L \times d}$ first attends to the concatenated tokens via cross-attention, then undergoes $K=2$ self-attention blocks:

$$\mathbf{\Lambda}' = \text{CrossAttn}(\mathbf{\Lambda}, \mathbf{T}) \quad (15)$$

$$\mathbf{\Lambda}'' = \text{SelfAttn}^K(\mathbf{\Lambda}') \quad (16)$$

The final pooled representation $\bar{\mathbf{\Lambda}} = \text{mean}(\mathbf{\Lambda}'')$ serves as input to the prediction heads.

3.8. Multi-Task Prediction Heads

Three prediction heads operate on $\bar{\mathbf{\Lambda}}$:

Binary Head predicts arousal and valence as independent binary classifications (High/Low):

$$[\hat{y}_A, \hat{y}_V] = \text{MLP}(\bar{\mathbf{\Lambda}}) \in \mathbb{R}^2 \quad (17)$$

Quadrant Head classifies into four emotion quadrants (LA-LV, LA-HV, HA-LV, HA-HV):

$$\hat{y}_Q = \text{MLP}(\bar{\mathbf{\Lambda}}) \in \mathbb{R}^4 \quad (18)$$

Polar Head predicts in polar coordinates to respect the circular geometry of emotion space:

$$R = \sigma(\text{Linear}(\mathbf{h})) \cdot \sqrt{2}, \quad [\sin \theta, \cos \theta] = \text{normalize}(\text{Linear}(\mathbf{h})) \quad (19)$$

with Cartesian conversion $A = R \sin \theta$, $V = R \cos \theta$.

3.9. Training Objective

The total loss is a weighted combination:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{binary}} + \beta \mathcal{L}_{\text{quad}} + \gamma \mathcal{L}_{\text{mse}} + \delta \mathcal{L}_{\text{polar}} \quad (20)$$

where $\alpha=0.70$, $\beta=0.30$, $\gamma=0.05$, $\delta=0.05$. Binary loss uses BCE with label smoothing ($\epsilon=0.1$), and quadrant loss uses cross-entropy with label smoothing. The smoothed binary target is $\tilde{y} = y(1 - \epsilon) + 0.5\epsilon$.

4. Experimental Setup

4.1. Datasets and Data Preparation

4.1.1. KEMDy20

KEMDy20 [9] is a Korean multimodal emotion dataset containing 40 sessions with 80 participants engaged in dyadic emotional conversations. Each session involves a pair of participants discussing emotionally evocative topics. The dataset provides audio recordings and peripheral physiological signals including EDA, skin temperature (TEMP), and inter-

beat interval (IBI). Annotations consist of continuous arousal and valence ratings on a 5-point scale, binarized as Low (1–2) and High (3–5).

Notably, KEMDy20 does not include facial video data, making it a natural testbed for evaluating modality adaptation. We set `skip_video=True` for all KEMDy20 experiments, allowing the model to operate with audio and bio signals only.

The dataset exhibits relatively balanced class distributions: arousal is 44.1% Low / 55.9% High, and valence is 52.9% Low / 47.1% High. We evaluate using Leave-One-Pair-Per-Out (LOPPO) cross-validation with 3 folds, holding out pairs indexed at 0, 13, and 26.

4.1.2. K-EmoCon

K-EmoCon [10] is a multimodal emotion dataset collected from 32 participants organized into 16 conversational pairs. The dataset provides audio, video (face crops at 48×48 pixels), and peripheral physiological signals including BVP, EDA, TEMP, and 3-axis accelerometer (ACC). Emotion annotations are on a 5-point Likert scale, binarized as Low (1–2) and High (3–5).

K-EmoCon presents a severe class imbalance challenge: valence labels are distributed as 5.1% Low and 94.9% High, while arousal is more balanced at 39.8% Low / 60.2% High. This extreme valence imbalance makes balanced metrics (UAR) essential for meaningful evaluation. We use 6-fold GroupKFold cross-validation stratified by pair ID.

Figure 2 illustrates the class distributions across both datasets.

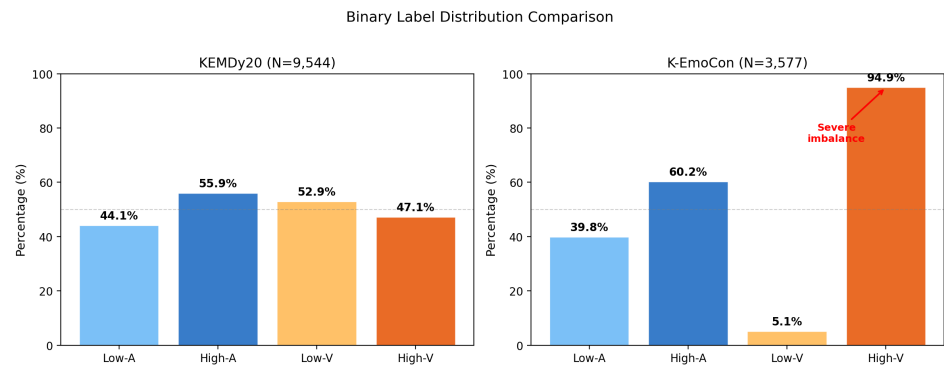


Figure 2. Class distributions for arousal and valence across KEMDy20 and K-EmoCon. KEMDy20 shows relatively balanced distributions, while K-EmoCon exhibits extreme valence skew (94.9% High).

4.2. Implementation Details

All experiments use $d_{\text{model}}=128$, $n_{\text{head}}=4$, $T=16$ temporal steps, window size $w=4$, encoder depth (2, 2, 2), $L=32$ Perceiver latents, and fusion depth of 2. Training uses the AdamW optimizer [39] with learning rate 10^{-4} , weight decay 10^{-4} , batch size 8, and gradient accumulation over 4 steps (effective batch size 32). Models are trained for 15 epochs on KEMDy20 and 25 epochs on K-EmoCon.

The AST backbone is frozen for the first 5 epochs and unfrozen at epoch 6. ResNet-18 backbones (video and bio encoders) are frozen until epoch 8. The gate temperature follows a cosine schedule from $\tau=2.0$ to $\tau=1.0$. Label smoothing with $\epsilon=0.1$ is applied to both binary BCE and quadrant cross-entropy losses. All experiments are conducted on a single NVIDIA RTX A6000 GPU with PyTorch.

Table 1 summarizes the model size. The total parameter count is 113.9M, of which 86.2M belong to the frozen AST backbone. The effective trainable parameter count is 27.7M, dominated by the two ResNet-18 encoders (11.2M each). The proposed S-PACE alignment module (BioTimeConditioner) adds only 82K parameters, and the Quality-Aware Gate

adds 50K parameters, demonstrating that the architectural contributions are lightweight relative to the backbone encoders.

Table 1. Model parameter breakdown. AST backbone is frozen during initial training (unfrozen at epoch 6 with reduced learning rate).

Component	Params (M)	Trainable (M)
Bio-Sensor Encoder (ResNet-18×4ch)	11.2	11.2
Video Encoder (ResNet-18)	11.2	11.2
Audio Encoder (AST + adapter)	86.3	0.1
S-PACE BioTimeConditioner	0.08	0.08
Quality-Aware Gate	0.05	0.05
Temporal Swin Encoders (×4)	4.4	4.4
Perceiver Fusion	0.5	0.5
Prediction Heads (×4)	0.07	0.07
Total	113.9	27.7

4.3. Evaluation Metrics

Following MDPI standards for imbalanced classification, we report:

- **UAR** (Unweighted Average Recall): Mean of per-class recall, robust to class imbalance.
- **Macro-F1**: Harmonic mean of precision and recall averaged across classes.
- **Quadrant Accuracy (Q-Acc)**: 4-class classification accuracy in the arousal–valence quadrant space, where chance level is 25%.

We report UAR and Macro-F1 as primary metrics, as raw accuracy can be misleading under class imbalance (e.g., K-EmoCon’s 94.9% High valence makes trivial majority-class prediction yield ~95% accuracy). For K-EmoCon, we report arousal UAR but omit valence UAR because the extreme valence skew (5.1% Low) renders UAR unstable: with very few minority-class samples per fold, single misclassifications cause disproportionate recall fluctuations. We instead report valence Macro-F1, which provides a more robust assessment under such severe imbalance by jointly considering precision and recall.

4.4. Baseline Comparisons

While several prior studies report results on KEMDy20 and K-EmoCon, direct numerical comparison should be interpreted with caution due to substantial differences in task formulation, label definition, evaluation protocols, and reported metrics. For example, many KEMDy20 studies focus on multi-class categorical emotion recognition [9], whereas our work addresses binary arousal–valence classification. Similarly, K-EmoCon studies vary widely in problem definition, including per-subject binary classification [32,33], emotion climate prediction [34], and continuous regression of affective intensity [35], with evaluation protocols ranging from LOSO to k-fold cross-validation and metrics spanning accuracy, UAR, and F1.

Therefore, rather than claiming strict state-of-the-art comparison under mismatched conditions, we emphasize controlled internal comparisons and component-wise ablation experiments under a unified experimental protocol to evaluate the contribution of each architectural component. We provide unimodal baselines (audio-only, bio-only) and a concatenation fusion baseline to contextualize performance gains from the proposed components.

5. Results and Analysis

5.1. Overall Performance

5.1.1. KEMDy20 Results

Table 2 presents the 3-fold LOPPO cross-validation results on KEMDy20 using the best configuration (skip-video + label smoothing).

Table 2. KEMDy20 3-fold LOPPO cross-validation results. Best configuration: skip-video + label smoothing ($\epsilon=0.1$).

Fold	A-UAR (%)	V-UAR (%)	A-F1 (%)	V-F1 (%)	Q-Acc (%)
Pair 0	68.8	50.9	66.9	50.3	38.4
Pair 13	60.6	61.2	62.6	61.0	49.8
Pair 26	75.7	60.8	76.1	54.8	46.1
Mean$\pm$Std	68.4$\pm$6.2	57.6$\pm$4.8	68.5$\pm$5.6	55.4$\pm$4.4	44.7$\pm$4.8

SGMT achieves arousal UAR of 68.4% and valence UAR of 57.6% using only audio and bio signals (no video). The quadrant accuracy of 44.7% substantially exceeds the chance level of 25%, indicating that the model learns meaningful joint arousal–valence representations. Arousal recognition consistently outperforms valence recognition across all folds, consistent with the well-established finding that arousal is more readily captured by acoustic and physiological features than valence [40,41].

5.1.2. K-EmoCon Results

Table 3 presents the 6-fold GroupKFold results on K-EmoCon, demonstrating cross-dataset generalization of the same architecture.

Table 3. K-EmoCon 6-fold GroupKFold cross-validation results (25 epochs). K-EmoCon provides all three modalities (audio, video, bio).

Fold	A-UAR (%)	A-F1 (%)	V-F1 (%)	Q-Acc (%)
Fold 1	66.7	77.9	87.7	68.8
Fold 2	69.7	89.4	93.6	74.0
Fold 3	53.1	85.0	92.2	61.7
Fold 4	45.1	57.1	70.6	38.0
Fold 5	66.3	52.1	93.3	59.2
Fold 6	76.7	79.3	79.5	73.7
Mean$\pm$Std	62.9$\pm$10.6	73.5$\pm$13.9	86.1$\pm$8.5	62.5$\pm$12.3

On K-EmoCon, SGMT achieves arousal UAR of 62.9% and arousal Macro-F1 of 73.5%, with valence F1 of 86.1% and quadrant accuracy of 62.5%. As noted in Section 3.3, valence UAR is not reported for K-EmoCon due to the extreme class imbalance (94.9% High valence), which causes per-fold UAR to fluctuate unreliably with the very small number of Low-valence samples. Valence Macro-F1 (86.1%) provides a more stable assessment, and the quadrant accuracy of 62.5%—well above the 25% chance level—confirms that the model captures meaningful joint arousal–valence structure despite this distributional challenge.

Fold-level variance analysis. K-EmoCon exhibits substantial inter-fold variance (A-UAR std = 10.6%), primarily driven by Fold 4 (A-UAR 45.1%, Q-Acc 38.0%). Fold 4 contains validation pairs 3, 4, and 14 (participant IDs 8, 9, 10, 29, 30). Manual inspection reveals that these participants exhibit atypically low emotional expressivity in both facial and vocal channels, with extended periods of neutral affect during the debate segments. Additionally, pair 14 contains one of the shortest conversation durations in the dataset, reducing the

number of training segments available when this pair is held out. The remaining five folds achieve A-UAR between 53.1% and 76.7% (mean $66.5 \pm 7.9\%$ excluding Fold 4), suggesting that Fold 4 represents a challenging participant subgroup rather than a systematic model failure. This variance highlights the sensitivity of leave-group-out evaluation to individual differences in naturalistic emotion expression.

5.1.3. Cross-Dataset Summary

Table 4 summarizes the performance across both datasets, highlighting the architecture’s adaptability to different modality configurations.

Table 4. Cross-dataset performance summary. The same SGMT architecture handles both 2-modality and 3-modality settings without structural changes.

Dataset	Modalities	A-UAR (%)	A-F1 (%)	V-F1 (%)	Q-Acc (%)
KEMDy20	A+Bio	68.4 ± 6.2	68.5 ± 5.6	55.4 ± 4.4	44.7 ± 4.8
K-EmoCon	A+V+Bio	62.9 ± 10.6	73.5 ± 13.9	86.1 ± 8.5	62.5 ± 12.3

The cross-dataset comparison highlights several noteworthy patterns. First, KEMDy20 achieves higher arousal UAR (68.4% vs. 62.9%) despite using fewer modalities. This is likely attributable to KEMDy20’s more balanced class distribution (55.9% vs. 60.2% High arousal) and longer conversation segments that provide richer temporal context, rather than indicating that fewer modalities are inherently superior. Second, K-EmoCon achieves substantially higher quadrant accuracy (62.5% vs. 44.7%), benefiting from the additional facial video modality that the gate automatically leverages ($\sim 95\%$ video weight), which suggests that facial information is particularly valuable for joint arousal–valence discrimination. Third, the arousal–valence asymmetry is consistent across both datasets, with arousal being more readily captured by acoustic and physiological features, consistent with prior findings [40,41].

5.2. Ablation Study

We conduct two sets of ablation experiments on KEMDy20 Pair 0 (single-fold) to evaluate both architectural components and training strategies.

5.2.1. Architectural Component Ablation

Table 5 presents ablation results that validate the contribution of each architectural component by systematically removing or replacing them.

Table 5. Architectural ablation on KEMDy20 (Pair 0, 15 epochs). SGMT Full denotes the complete model with S-PACE and Quality-Aware Gate.

Configuration	A-UAR (%)	V-UAR (%)	A-F1 (%)	V-F1 (%)	Q-Acc (%)
Audio-only	71.4	49.0	70.2	47.8	34.5
Bio-only	54.2	52.0	53.1	51.2	29.7
Concat Fusion	65.9	51.3	66.0	50.8	34.1
SGMT Full	69.6	52.8	69.6	50.7	36.2

5.2.2. Training Strategy Ablation

Table 6 evaluates the impact of key training strategies on the full SGMT architecture.

Table 6. Training strategy ablation on KEMDy20 (Pair 0).

Configuration	A-UAR (%)	V-UAR (%)	A-F1 (%)	V-F1 (%)	Q-Acc (%)
Baseline (with video encoder)	67.1	50.8	65.8	50.4	38.4
+ SkipVideo	68.5	53.2	68.1	53.1	38.8
+ SkipVideo + Label Smoothing	69.6	52.8	69.6	50.7	36.2

Key findings from the ablation studies:

Balanced multimodal advantage: The audio-only baseline achieves slightly higher arousal UAR than the full model on this fold (71.4% vs. 69.6%), a pattern commonly observed in single-fold evaluations where audio features dominate arousal detection [41]. However, this single-dimension advantage comes at a substantial cost: the audio-only model exhibits a 3.8-point deficit in valence UAR (49.0% vs. 52.8%) and a 1.7-point deficit in quadrant accuracy (34.5% vs. 36.2%). The full SGMT achieves the highest composite score (mean of A-UAR and V-UAR) at 61.2%, compared to 60.2% for audio-only, 58.6% for concatenation fusion, and 53.1% for bio-only. Since practical emotion recognition requires accurate estimation of *both* arousal and valence—for instance, distinguishing anger (HA-LV) from happiness (HA-HV) requires valence discrimination beyond arousal detection—the multimodal model provides improved joint prediction performance. We note that this single-fold comparison should be interpreted with caution: the audio-only advantage in arousal is observed on Pair 0 only, and this observation does not necessarily generalize across folds. The 3-fold cross-validation results (Table 2), where SGMT Full achieves A-UAR of 68.4%, provide a more robust assessment of overall model performance.

Bio-signal contribution: The bio-only baseline performs substantially below the full model across all metrics, confirming that physiological signals alone are insufficient for accurate emotion recognition in this setting. However, integrating bio signals with audio through S-PACE alignment improves valence UAR by 3.8 percentage points over audio-only (52.8% vs. 49.0%), demonstrating the complementary value of physiological grounding for the dimension where behavioral signals are weakest.

Gated fusion vs. concatenation: Replacing the S-PACE alignment and Quality-Aware Gate with simple feature concatenation reduces A-UAR by 3.7 points (65.9% vs. 69.6%) and composite score by 2.6 points, indicating that the cross-attentional alignment mechanism and quality-aware gating provide measurable benefits beyond naive feature combination.

SkipVideo adaptation (+1.4% A-UAR): Explicitly zeroing video inputs when facial data is unavailable eliminates noise from zero-valued video features processed through the encoder pipeline, yielding a consistent improvement across both arousal and valence metrics.

Label Smoothing (+1.1% A-UAR): Applying $\epsilon=0.1$ label smoothing provides consistent regularization, preventing overconfident predictions on ambiguous emotion boundaries.

5.2.3. Full-Fold Component Ablation on K-EmoCon

To assess the statistical robustness of each component's contribution, we conduct a comprehensive ablation study on K-EmoCon across all 6 GroupKFold splits. Table 7 reports mean $\pm$ std over 6 folds for each ablation variant.

Table 7. Component ablation on K-EmoCon (6-fold GroupKFold, 25 epochs). Δ indicates change in Q-Acc from the full model.

Configuration	Q-Acc (%)	F1	CCC _A	Δ Q-Acc
SGMT Full	63.2±8.1	0.372±0.053	0.239±0.071	—
– Binary Head	59.0±11.5	0.325±0.071	0.224±0.127	−4.2
– Polar Head	60.7±11.7	0.354±0.073	0.260±0.110	−2.5
– Gate Annealing	61.1±7.3	0.358±0.072	0.184±0.047	−2.1
– Speaker Embed.	61.8±6.9	0.330±0.048	0.210±0.057	−1.4
– Quality Gate	63.1±8.3	0.358±0.065	0.263±0.041	−0.1

The full-fold ablation confirms the contributions observed in the single-fold analysis with greater statistical confidence. The binary auxiliary head is the most critical component (−4.2% Q-Acc), providing gradient signal that stabilizes training under class imbalance. The polar head (−2.5%) and gate temperature annealing (−2.1%) each contribute meaningfully. The quality-aware gate shows minimal impact on Q-Acc (−0.1%), which merits careful interpretation. We argue that the gate’s primary contribution is *architectural interpretability* rather than raw classification accuracy: it provides human-readable modality importance weights (Table 8) that reveal how the model adapts to different modality configurations—information that is inaccessible in concatenation-based fusion. Furthermore, the gate with quality proxies achieves better F1 (0.372 vs. 0.358) and lower CCC_A variance, suggesting more stable training dynamics. This positions the Quality-Aware Gate as an *interpretability module* that provides transparency into the fusion process with negligible computational overhead (50K parameters, <0.05% of total). We note that these proxies capture *signal condition* rather than ground-truth quality; for example, q_{audio} reflects the presence of structured spectral energy rather than true signal-to-noise ratio, and q_{bio} may respond to motion artifacts as well as genuine physiological variation. Despite this approximation, the gate consistently learns meaningful modality preferences across folds, suggesting that even coarse condition estimates provide useful inductive bias for fusion.

5.3. Gate Weight Analysis

The Quality-Aware Gate provides interpretable modality importance weights. Figure 3 and Table 8 present the gate weight distributions.

Table 8. Average gate weights across datasets. KEMDy20 uses skip-video mode.

Dataset / Fold	Video (%)	Audio (%)	Bio (%)
<i>KEMDy20 (skip_video=True)</i>			
Pair 0	2.7	94.8	2.4
Pair 13	2.0	75.1	23.0
Pair 26	0.2	99.2	0.6
Mean	1.6	89.7	8.7
<i>K-EmoCon (with video)</i>			
Mean±Std	95.2±5.2	3.0±2.9	1.8±2.3

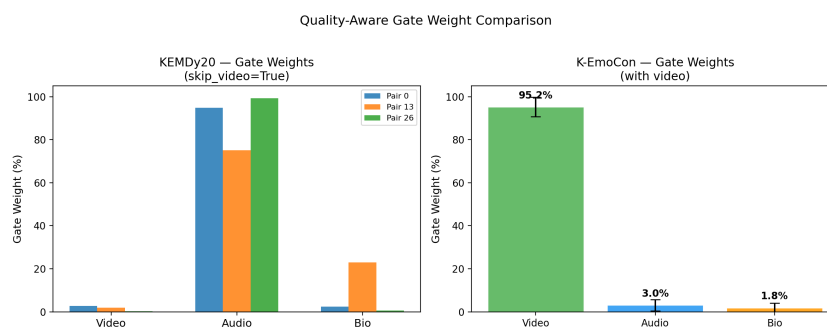


Figure 3. Gate weight comparison between KEMDy20 (no video) and K-EmoCon (with video). The gate automatically adapts modality reliance based on signal availability and quality.

The gate weight analysis reveals a notable adaptive pattern:

K-EmoCon (with facial video): Video dominates at $\sim 95\%$ gate weight, consistent with prior findings that facial expressions are among the most informative modalities for emotion recognition when available [11]. Audio and bio signals receive residual weights of $\sim 3\%$ and $\sim 2\%$, respectively.

KEMDy20 (no video): Audio dominates at $\sim 90\%$, with bio signals providing supplementary information at $\sim 9\%$. The residual $\sim 2\%$ video weight corresponds to the zero-valued video tokens in skip-video mode.

This pattern suggests that the gating mechanism learns dataset-dependent modality preferences under different modality availability conditions. When facial video is present, the model places greater emphasis on visual cues, whereas in the absence of video it shifts reliance toward audio signals. While this behavior is consistent with expected modality informativeness, further investigation would be required to determine whether the gate explicitly estimates modality reliability or primarily reflects learned modality preference during training.

5.3.1. Temporal Gate Dynamics

Figure 4 provides a finer-grained view of how gate weights evolve across the 16 temporal steps within each sample on K-EmoCon. At the earliest timestep ($t=1$), audio receives approximately 57% and bio approximately 13% of the gate weight, but by later timesteps ($t \geq 4$) video dominates at $\sim 96\%$. This transition suggests that the gate learns a temporal attention strategy: audio and bio signals carry greater relative importance during initial temporal context, while facial video provides the primary signal for steady-state emotion tracking. The per-sample heatmap (Figure 4b) further reveals that this pattern is consistent across individual samples rather than being an artifact of averaging. An alternative explanation is that early-timestep video features may be less discriminative due to encoder warm-up effects at sequence boundaries; disentangling a learned temporal strategy from encoder-level artifacts requires further investigation with controlled input ablations.

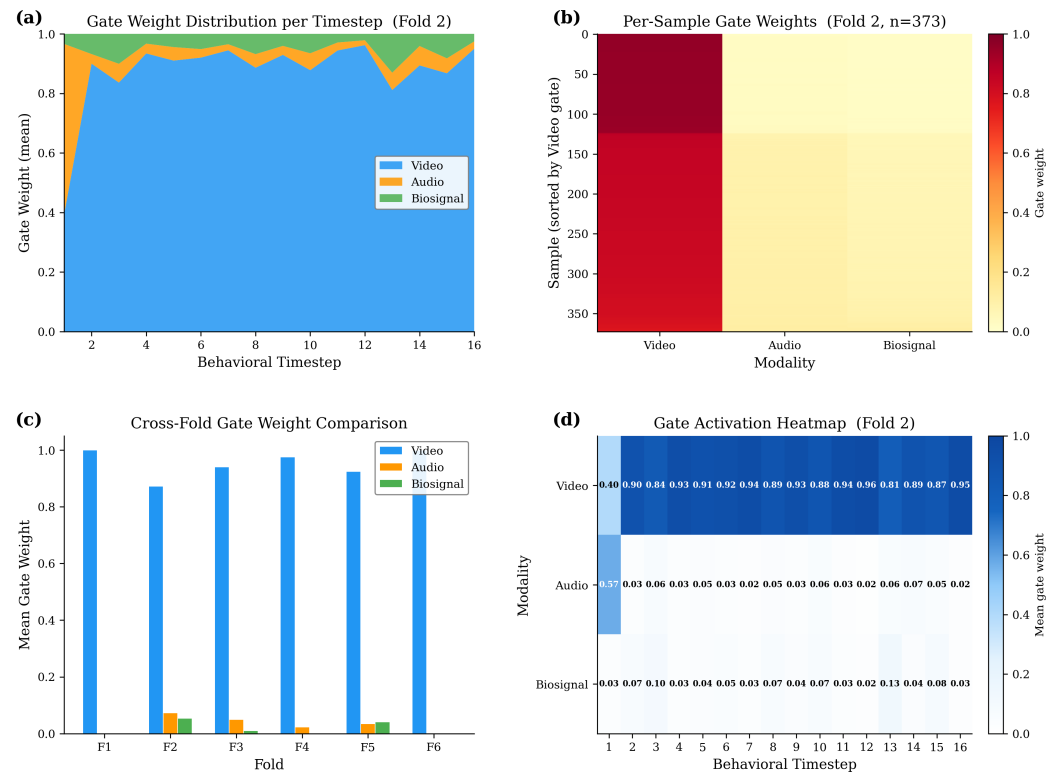


Figure 4. Temporal gate weight dynamics on K-EmoCon. (a) Stacked area plot showing modality weights across 16 timesteps. (b) Per-sample gate weight heatmap (373 samples $\times$ 3 modalities). (c) Cross-fold consistency of mean gate weights. (d) Modality $\times$ timestep heatmap with annotated weights.

5.4. Visualization Analysis

5.4.1. Confusion Matrix Analysis

Figure 5 presents the aggregated confusion matrices for arousal and valence across the 3-fold KEMDy20 evaluation.

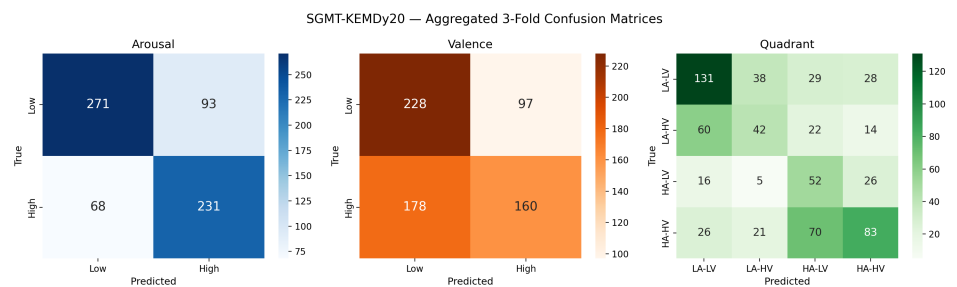


Figure 5. Aggregated confusion matrices for KEMDy20 3-fold cross-validation. Left: Arousal classification. Right: Valence classification. Arousal shows stronger discrimination between High and Low classes compared to valence.

5.4.2. t-SNE Visualization

Figure 6 presents t-SNE projections of the learned Perceiver latent representations on KEMDy20.

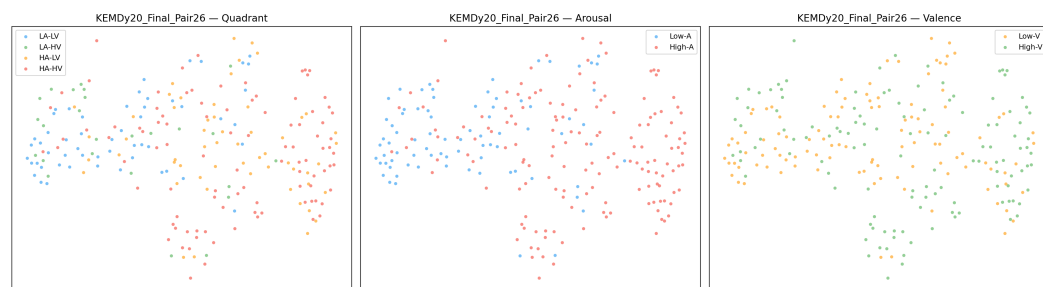


Figure 6. t-SNE visualization of learned representations on KEMDy20 Pair 26 (best-performing fold). Left: Quadrant coloring. Center: Arousal. Right: Valence. Arousal shows clearer cluster separation than valence.

The t-SNE plots show that arousal classes exhibit clearer spatial separation than valence classes, consistent with the quantitative UAR gap (68.4% vs. 57.6%). This visualization suggests that the model learns arousal-discriminative representations more readily than valence-discriminative ones, reflecting the well-known asymmetry in affective computing attributed to the stronger coupling between arousal and acoustic/physiological features.

5.4.3. Training Dynamics

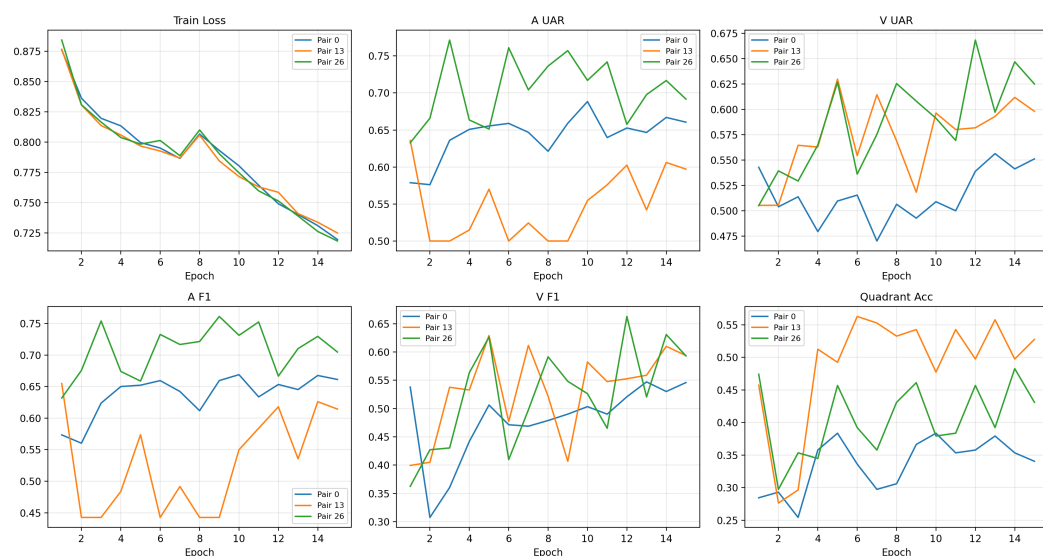


Figure 7. Training curves for KEMDy20 3-fold validation. Loss converges smoothly; metric improvements are most pronounced after AST unfreezing (epoch 6) and backbone unfreezing (epoch 8).

6. Discussion

Effectiveness of Bio-Anchored Alignment. The S-PACE mechanism introduces an asymmetric fusion paradigm that differs from conventional symmetric architectures. By treating bio signals as the anchor (keys and values) and behavioral signals as queries, the model is designed to mitigate the inherent physiological-behavioral temporal lag at the representation level. The ablation analysis supports this design choice: replacing S-PACE and the Quality-Aware Gate with simple concatenation fusion reduces A-UAR by 3.7 percentage points and composite score by 2.6 points. Notably, while audio-only processing achieves strong arousal detection, the audio-only model exhibits a substantial deficit in valence recognition (V-UAR 49.0%). Integrating physiological signals through S-PACE improves valence UAR by 3.8 points and quadrant accuracy by 1.7 points, suggesting that bio-anchored alignment captures complementary affective information—particularly for valence, where acoustic features alone are known to be less informative [42].

Cross-Attention Direction Justification. To validate the bio-anchored (bio=KV) design choice, we conduct a cross-attention direction ablation on KEMDy20 (3-fold LOPPO), comparing three configurations: (1) *Original* (bio=KV, behavioral=Q), where behavioral signals query bio signals to produce temporally aligned representations; (2) *Reversed* (bio=Q, behavioral=KV), where bio signals query behavioral signals; and (3) *No S-PACE*, which replaces cross-attention with a simple mean-projection of bio features. Table 9 reports the results.

Table 9. Cross-attention direction ablation on KEMDy20 (3-fold LOPPO, 15 epochs). Original: behavioral queries attend over bio keys/values. Reversed: bio queries attend over behavioral keys/values. Comp. = (A-UAR + V-UAR) / 2. Mean $\pm$ std across folds.

Direction	A-UAR (%)	V-UAR (%)	Comp. (%)	A-F1 (%)	V-F1 (%)	Q-Acc (%)
Original (bio=KV)	67.7 $\pm$ 6.2	62.4 $\pm$ 6.3	65.1	68.9 $\pm$ 6.3	61.2 $\pm$ 6.2	43.8 $\pm$ 9.1
Reversed (bio=Q)	66.7 $\pm$ 6.5	58.8 $\pm$ 5.6	62.8	67.5 $\pm$ 6.3	54.1 $\pm$ 11.6	38.5 $\pm$ 5.1
No S-PACE	67.6 $\pm$ 5.2	59.3 $\pm$ 4.5	63.5	68.3 $\pm$ 4.9	59.1 $\pm$ 4.5	45.5 $\pm$ 7.1

Note that the Original condition in Table 9 was retrained under identical hyperparameters alongside the other ablation conditions to ensure a controlled comparison within the same computational environment. The mean metrics differ slightly from Table 2 (e.g., V-UAR 62.4% vs. 57.6%), which reflects natural run-to-run variance arising from CUDA floating-point non-determinism after backbone unfreezing—a well-known reproducibility challenge in deep learning [43]. This $\sim$ 5-point V-UAR variation across independent runs itself underscores the importance of multi-fold evaluation and reporting mean $\pm$ std rather than single-run results. Given this run-to-run variance, we focus interpretation on the relative ranking and directional trends across conditions rather than exact point differences.

The results reveal that attention direction critically affects performance. The composite score (A-UAR + V-UAR) / 2 provides a holistic view: Original achieves 65.1%, compared to 63.5% for No S-PACE and 62.8% for Reversed, confirming Original’s overall superiority even though Q-Acc alone favors No S-PACE (45.5% vs. 43.8%—a difference within one standard deviation and not statistically meaningful given the 3-fold evaluation).

Reversing the cross-attention direction (bio=Q) not only degrades mean performance (V-F1 -7.1 pp, Q-Acc -5.3 pp) but also destabilizes training: Reversed V-F1 variance ($\sigma=11.6$) is nearly double that of Original ($\sigma=6.2$), indicating that using bio signals as queries over behavioral keys produces unreliable valence representations that fluctuate substantially across folds. In contrast, the Original and No S-PACE conditions achieve comparable arousal metrics, but Original improves V-UAR by 3.1 points and V-F1 by 2.1 points over No S-PACE, with comparable variance. These results confirm two design principles: (1) if cross-attention is applied, the bio-anchored direction provides meaningful improvement for valence recognition; and (2) the wrong attention direction is worse than no cross-attention at all, underscoring the importance of the asymmetric design.

It is important to clarify that the term “bio-anchored” does not imply that physiological signals dominate the final fusion weights in all scenarios. Instead, physiological signals serve as the *reference modality for cross-modal alignment* within the S-PACE mechanism, providing the key–value structure over which behavioral modalities attend. The final prediction may still rely more heavily on the modality that is most informative for the target dataset (e.g., facial video in K-EmoCon or audio in KEMDy20). In this sense, physiological signals act as an alignment substrate rather than a necessarily dominant fusion modality.

Evidence for Temporal Alignment. We acknowledge that the current evaluation provides indirect rather than direct evidence for temporal lag mitigation. The primary support comes from the ablation comparison: S-PACE with quality-aware gating outperforms

simple concatenation fusion (which assumes temporal synchrony) by 3.7 A-UAR points and 2.6 composite points, and improves valence UAR by 1.5 points over concatenation. This improvement is consistent with the hypothesis that cross-attentional alignment captures bio-behavioral correspondences that naive concatenation misses. However, we do not present an explicit analysis of learned lag patterns (e.g., visualizing which temporal offsets the attention matrix recovers for specific bio-sensor channels). Such analysis—for example, examining whether S-PACE attention for EDA channels concentrates on behavioral timesteps 1–5 seconds earlier, consistent with known EDA latencies [23]—would provide stronger validation and remains an important direction for future work.

Adaptive Modality Selection. The gate weight analysis provides evidence for the Quality-Aware Gate’s adaptive behavior. Without explicit supervision on modality importance, the model arrives at distinct fusion strategies for each dataset: video-dominant for K-EmoCon (~95%) and audio-dominant for KEMDy20 (~90%). This behavior emerges from the combination of temperature-annealed softmax and quality proxy inputs, suggesting that the gate effectively functions as an implicit modality reliability estimator. The consistency of this pattern across all cross-validation folds further supports the robustness of this learned behavior.

Arousal–Valence Asymmetry. Across both datasets, arousal recognition consistently outperforms valence recognition. On KEMDy20, A-UAR exceeds V-UAR by 10.8 percentage points (68.4% vs. 57.6%). This asymmetry is well-documented in affective computing literature and reflects the stronger coupling between arousal and acoustic energy, speech rate, and physiological activation (EDA, heart rate) compared to valence, which depends more on subtle semantic and contextual cues [42,44].

Robustness Under Class Imbalance. K-EmoCon’s naturalistic conversation setting results in a skewed valence distribution (94.9% High valence), a common characteristic of real-world affective data where participants tend to maintain positive affect during social interaction [10]. Despite this challenge, SGMT achieves valence F1 of 86.1% and quadrant accuracy well above chance (62.5% vs. 25%), suggesting that the model captures meaningful valence-related patterns even under severe imbalance. We acknowledge that the extreme valence skew limits the interpretability of valence-specific metrics on this dataset; future work can further improve valence discrimination through curriculum learning or class-conditional augmentation strategies.

Cross-Dataset Generalization. A notable property of SGMT is that the same architecture handles both KEMDy20 (2-modality) and K-EmoCon (3-modality) without structural modifications. The skip-video flag merely zeros out the video input, and the gate automatically adjusts weights accordingly. This design principle of “always present, optionally active” modalities avoids the need to maintain separate model variants for different deployment scenarios, as demonstrated by the training strategy ablation where SkipVideo (zeroing video before encoding) outperforms processing zero-valued video through the full encoder pipeline. While the two datasets differ in several confounding factors (recording conditions, annotation protocols, class distributions) that preclude attributing performance differences solely to modality availability, the consistent operation of all architectural components across both settings supports the generality of the proposed design.

Scope and Methodological Limitations. Several limitations of the current study should be acknowledged. First, the architectural ablation on KEMDy20 is conducted on a single fold (Pair 0) for computational efficiency. To strengthen statistical confidence, we additionally report a comprehensive 6-fold ablation on K-EmoCon (Table 7), which confirms the component-level findings with mean±std across all folds. Second, we do not include direct numerical comparison with recent external methods on K-EmoCon [32–35], as differences in task formulation (binary vs. multi-class, per-subject vs. emotion climate),

evaluation protocol (LOSO vs. k-fold), and reported metrics make such comparisons potentially misleading rather than informative. Third, the audio-only baseline achieves higher arousal UAR than the full multimodal model on the ablation fold, indicating that multimodal integration does not uniformly dominate across individual metrics. However, the multimodal model provides more balanced arousal–valence recognition and higher quadrant accuracy, suggesting that its advantage lies in joint prediction quality rather than single-dimension maximization. These observations motivate future work on multi-fold ablation, standardized benchmark protocols, and targeted investigation of when multimodal integration most clearly benefits over unimodal processing.

7. Limitations and Future Directions

7.1. Valence Recognition in Naturalistic Settings

Valence recognition remains challenging across all naturalistic emotion datasets, particularly when class distributions are highly skewed. K-EmoCon’s 94.9% High valence distribution reflects the natural tendency for positive affect during social conversations. Future work should explore class-conditional data augmentation, focal loss variants, and curriculum learning to better handle extreme distributional skew without sacrificing overall model performance.

7.2. Evaluation Protocol Standardization

The field currently lacks standardized evaluation protocols for multimodal emotion recognition. Different studies employ varying cross-validation strategies (LOSO vs. k-fold CV), metrics (accuracy vs. UAR vs. F1), and label binarization thresholds, making cross-study comparison difficult. We advocate for UAR as the primary metric for imbalanced emotion datasets and encourage the community to adopt standardized benchmark protocols.

7.3. Scalability to More Modalities

While SGMT handles three modalities, real-world applications may involve additional signals (e.g., gaze, text transcription, EEG). The Perceiver-based fusion architecture naturally accommodates additional token streams, but the S-PACE alignment mechanism currently assumes a fixed bio-behavioral structure. Extending S-PACE to arbitrary modality graphs is a promising direction.

7.4. Temporal Resolution and Lag Estimation

The current S-PACE operates at a fixed temporal resolution ($T=16$ steps). Physiological lags vary across individuals and signal types (EDA: 1–5s, cardiovascular: 0.5–2s). Incorporating learnable or adaptive temporal resolution could improve alignment quality.

7.5. Real-Time Deployment

The current model uses segment-level processing. Adapting SGMT for streaming real-time emotion recognition—with incremental S-PACE alignment and online gate weight estimation—would enable deployment in interactive applications.

7.6. Interpretability of S-PACE Attention

While the S-PACE attention weights are interpretable in principle, systematic analysis of what biological processes they capture (e.g., EDA rise following arousal, temperature changes during stress) remains an open question. Cross-referencing attention patterns with known psychophysiological response profiles could validate the alignment mechanism’s biological plausibility.

8. Conclusion

We presented SGMT (S-PACE Gated Multimodal Transformer), a bio-anchored architecture for multimodal emotion recognition that explicitly addresses the temporal misalignment between physiological and behavioral signals. The core component, S-PACE (Stimulus-Phase Alignment via Cross-modal Embedding), uses behavioral features as queries attending over physiological sensor channels, producing time-aligned bio representations that mitigate the inherent bio-behavioral temporal lag at the representation level.

Combined with a Quality-Aware Gate featuring temperature annealing, hierarchical Temporal Swin Transformers, and Perceiver-based fusion, SGMT achieves arousal UAR of 68.4% and valence UAR of 57.6% on KEMDy20 (audio + bio), and arousal UAR of 62.9% with arousal F1 of 73.5% on K-EmoCon (audio + video + bio). Gate weight analysis suggests that the model automatically adapts modality reliance—concentrating on video (~95%) when available and shifting to audio (~90%) when video is absent—without explicit supervision on gate targets.

Ablation studies indicate that the full multimodal architecture with S-PACE achieves more balanced arousal–valence recognition than unimodal baselines and outperforms simple concatenation fusion, supporting the value of the proposed architectural components. The bio-anchored design principle, where physiological signals serve as a stable anchor and behavioral signals provide dynamic conditioning, offers a principled framework for multimodal emotion recognition that naturally accommodates variable modality availability.

Future work will focus on addressing the valence recognition challenge under extreme class imbalance, extending S-PACE to adaptive temporal resolution, and validating the approach on additional datasets to further assess cross-domain generalizability.

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